

something

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Step 1 — Load Data

- Read CSV into Spark DataFrame
 - Define schema manually (avoid automatic type inference for huge data)
 - Verify columns and data types
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Step 2 — Data Quality Check

Check:

- Missing values
 - Duplicate rows
 - Invalid timestamps
 - Invalid user IDs/session IDs
 - Invalid product IDs
 - Negative or impossible amounts
 - Unknown event types
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Step 3 — Label Preparation

Convert:

outcome none $\rightarrow$ 0 purchase $\rightarrow$ 1

Create your target column:

label

Step 4 — Remove Data Leakage

Check that no feature contains future information.

Example:

- Do not use “purchase event happened after browsing” as a feature.
 - Features must represent information available before prediction.
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Step 5 — Session-Level Aggregation

Your current data is event-level:

user → session → many events

Convert it into:

one row = one session

Example:

Before:

session 101 page_view click product_view add_to_cart purchase

After:

session 101 total_events = 5 page_views = 1 clicks = 1 cart_adds = 1 products_viewed = 1 label = 1

This is the most important transformation.

Step 6 — Feature Extraction

Create predictive features.

From event type:

- Total number of events
- Number of page views
- Number of product views
- Number of clicks
- Number of add-to-cart events
- Number of login events

From session:

- Session duration
- First event time
- Last event time
- Number of unique products viewed
- Number of products added to cart

From product:

- Number of unique products
- Product category features (if available later)

From amount:

- Total amount viewed
- Average product amount
- Maximum product amount

Step 7 — Time-Based Features

Extract:

From timestamp:

- Hour of day
- Day of week
- Weekend/weekday
- Month/season

Example:

Purchase probability may differ:

2 AM vs 8 PM weekday vs weekend

Step 8 — Handle Categorical Features

Your string columns:

- event type
- product ID

Need conversion.

Methods:

- One-hot encoding
- Label encoding
- Target encoding
- Embedding (advanced)

For product ID, be careful because unique IDs can create millions of features.

Step 9 — Handle Class Imbalance

Your data:

No purchase: 70% Purchase: 30%

This is not extremely imbalanced, but check after session aggregation.

Options:

- Class weights
- Sampling
- Evaluation metrics like F1, AUC instead of only accuracy

Step 10 — Feature Vector Creation

Combine all features into one ML input:

features [event_count, session_duration, cart_add_count, hour, amount, ...]

Step 11 — Train/Test Split

Split data:

Example:

80% training 20% testing

For timestamp data, preferably use:

past → train future → test

not random split.

Step 12 — Train Models

Start with:

1. Logistic Regression (baseline)
 2. Random Forest
 3. Gradient Boosted Trees
 4. XGBoost/LightGBM (often strong for this type of data)
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Step 13 — Evaluate Model

Measure:

- Accuracy (not enough)
- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC

For purchase prediction, usually focus on:

- How many actual buyers you can detect
 - How many predictions are correct
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Step 14 — Tune Model

Adjust:

- Model parameters
- Feature selection
- Threshold probability

Example:

Default:

purchase probability > 0.5

Maybe business wants:

purchase probability > 0.3
to target more users.

Step 15 — Deploy Prediction Pipeline

Final flow:

New User Events | v Spark Processing | v Feature Extraction | v ML Model | v Purchase Probability

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